

The delay-area identity for a first-order all-pass

Claim

It is often stated that the amount of group-delay ripple a first-order all-pass section can equalize can be estimated by integrating the group-delay curve. This is true, and it is in fact an *exact* identity, not just a rule of thumb. Verified numerically (via this toolbox's `AnlzdH.m`) for several pole radii and angles, real and complex, to 6 decimal places.

Setup

A first-order real digital all-pass with pole at $z = r$ ($0 < r < 1$) has

$$H(z) = \frac{z^{-1} - r}{1 - rz^{-1}}$$

Its zero at $z = 1/r$ is the pole's mirror image outside the unit circle, which is what makes $|H(e^{j\omega})| = 1$ for all ω – an all-pass. Differentiating the phase (equivalently, using the exact pole/zero sum in `AnlzdH.m`) gives the standard closed form for the group delay:

$$\tau(\omega) = -\frac{d\phi}{d\omega} = \frac{1 - r^2}{1 - 2r \cos \omega + r^2}$$

This is the **Poisson kernel** $P_r(\omega)$ from classical harmonic analysis and complex analysis – the same kernel that solves the Dirichlet problem on the disk and appears in Fourier series summability theory. It satisfies the well-known normalization identity

$$\int_{-\pi}^{\pi} P_r(\omega) d\omega = \int_{-\pi}^{\pi} \frac{1 - r^2}{1 - 2r \cos \omega + r^2} d\omega = 2\pi \quad \text{for every } 0 \leq r < 1$$

so $\int_{-\pi}^{\pi} \tau(\omega) d\omega = 2\pi$ exactly, *independent of r* . Only the *shape* changes with r : small r gives a broad, shallow bump; $r \rightarrow 1$ concentrates the same fixed area into an ever-narrower, ever-taller spike (the Poisson kernel is the textbook example of an “approximate identity” converging to a Dirac delta as $r \rightarrow 1$).

A more general proof: winding number

The direct-integration proof above is elementary but doesn't explain *why*. Here is the reason, which also immediately generalizes to higher-order and complex-pole all-passes.

A first-order all-pass factor $B(z) = \frac{z - p}{1 - p^*z}$ (a **Blaschke factor**) is a Möbius transformation that maps the open unit disk bijectively onto itself. Restricted to the boundary $|z| = 1$, it is therefore a homeomorphism of the circle to itself – meaning as ω sweeps once around $-\pi \rightarrow \pi$, $H(e^{j\omega})$ traces the unit circle *exactly once*, in the same rotational sense. So the total phase change over one period is exactly

$$\phi(\pi) - \phi(-\pi) = -2\pi$$

(the argument principle / winding number, here trivial since it is degree 1). Since $\tau = -d\phi/d\omega$,

$$\int_{-\pi}^{\pi} \tau(\omega) d\omega = -[\phi(\pi) - \phi(-\pi)] = 2\pi$$

This is a **topological invariant** – it depends only on there being exactly one pole strictly inside the unit circle, not on where. This is why the closed-form integral above came out independent of r , and it immediately gives the general result for higher order: an N -th order all-pass (product of N Blaschke factors, or one N -pole all-pass) winds the unit circle N times, so

$$\int_{-\pi}^{\pi} \tau(\omega) d\omega = 2\pi N$$

Group delay adds linearly for a cascade, so this also follows directly from the first-order result by superposition.

In this toolbox’s own units

Everything in this toolbox (`plot_dGD`, `AnlzdH`) works in cyclic frequency $f = \omega/2\pi \in [-0.5, 0.5]$ rather than radians, where the identity becomes simply

$$\int_{-0.5}^{0.5} \tau(f) df = 1 \text{ sample, per pole}$$

exactly what the numeric test confirms: `trapz(f, gdH) = 1.000000` for every pole tried, real or complex, any radius or angle.

The practical rule of thumb

Each first-order all-pass section has a **fixed total delay “budget” of exactly 1 sample**, spread over the band however its pole radius/angle dictates – a larger radius concentrates more of that budget into a narrower, taller peak near the pole’s frequency, but can never exceed area 1 in total. So to estimate how many first-order all-pass sections a delay-equalizer needs to flatten a given group-delay ripple:

1. Take the ripple you want to cancel (the deviation of $\tau(\omega)$ from flat/constant, within the band that matters).
2. Compute its area, `trapz(f, ripple)`.
3. Divide by 1 (or by 2π if integrating in ω) – the result estimates the equalizer order (number of first-order sections) needed.

This is a genuine estimate, not exact, because it only tells you the required “budget” – it doesn’t guarantee a good *fit* to the ripple’s shape (a wide, shallow ripple vs. a narrow, tall one of the same area may need different pole placement/order to actually flatten well). But as an order-of-magnitude / lower-bound tool for how many all-pass sections you’ll need, it is exactly the area argument above, and is standard in classical filter-equalizer design texts (e.g. Zverev’s *Handbook of Filter Synthesis* uses this kind of delay-area reasoning for equalizer design). It is the same underlying identity – total phase winding = $2\pi \times (\text{pole count})$ – that guarantees this toolbox’s `LinPhFiltr/EqualFiltr` equalizers need roughly one all-pass section per unit of accumulated group-delay-ripple area they are asked to flatten.